XCITY AI COMPETITION · 1,656 KM² · SAN JUAN, ARGENTINA

Terrace Oasis: a city that grows its own shade beside the first fields



city-evening · ChatGPT image generation (chatgpt.com, web app, October 2026) from renders/city-evening.png · illustrates the geometry, not evidence

The scheme

Reference example, not an entry. The first city stands on the flat terrace directly east of the pilot area, between the fields and a 30 m bluff, facing the Andes. About 41,340 people live in low courtyard houses on a 300 m grid; every street is an acequia street, shaded by trees watered by gravity from a tank on the bluff; machines keep to their own 600 m grid. Across the site, four oasis cities hang on one corridor, the Line, from the RN 20 gate to the port in the east. The city gives up growth speed, not water.

41,340	**7,891**	**40**	**410**
residents, phase one	jobs in the city	MW of compute	MWp of PV
3,599,292	**100**	**36**	**30**
m³ of water a year	% hourly carbon-free	km² master design	human-machine interfaces

Key nodes: C01 Entrance and arrival sequence · A01 A future housing prototype · A03 Public space in heat and Zonda wind

Declared trade-off: Growth speed: the city does not grow faster than its measured water allows

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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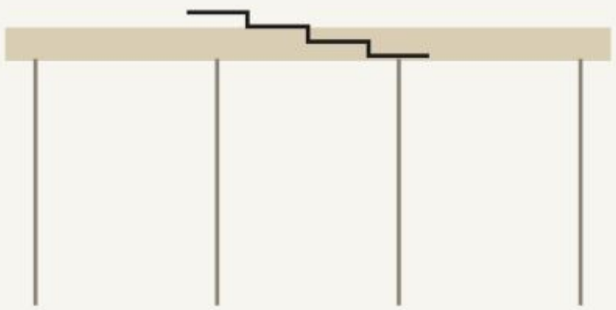
Agent task 1 · drawn by the agent with tools/make_pages.py



Terraza

the water comes down our street, and the street is ours

THE MARK IN USE



On a shade canopy, cut into the fascia



On a delivery robot's status light



Stamped on an olive-oil tin

NAMING · IN WATER ORDER

Terraza

the city

Avenida del Agua

the main street, from the tank on the bluff

Acequia 1, 2, 3 …

the acequia streets, numbered downhill

Acequia 4 · Patio 2

an address: how near you are to the water

Plaza del Agua

the main square

La Llegada

the arrival, under the coach canopy

COLOUR DOES A JOB

land and buildings

water and planting

machine territory

One sans-serif family in two weights. Signs sit on the ground, not on poles. Paving changes texture 20 m before every machine crossing.

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Terrace Oasis: a city that grows its own shade beside the first fields

Scale 1 · structural



Legend

- Reserve 899 km²
- Agri-production 642 km²
- Energy + compute 58 km²
- Industry + automation 30 km²
- Living + public life 21 km²
- Trade + port 3.30 km²
- Site boundary (1,656 km²)
- Pilot area - fixed, not designed
- Master design area
- Seasonal channels (DEM-derived)
- Key nodes

Why the structure sits where it does

Reference example, not an entry. The first city stands on the flat terrace directly east of the pilot area, between the fields and a 30 m bluff, facing the Andes. About 41,340 people live in low courtyard houses on a 300 m grid; every street is an acequia street, shaded by trees watered by gravity from a tank on the bluff; machines keep to their own 600 m grid. Across the site, four oasis cities hang on one corridor, the Line, from the RN 20 gate to the port in the east. The city gives up growth speed, not water.

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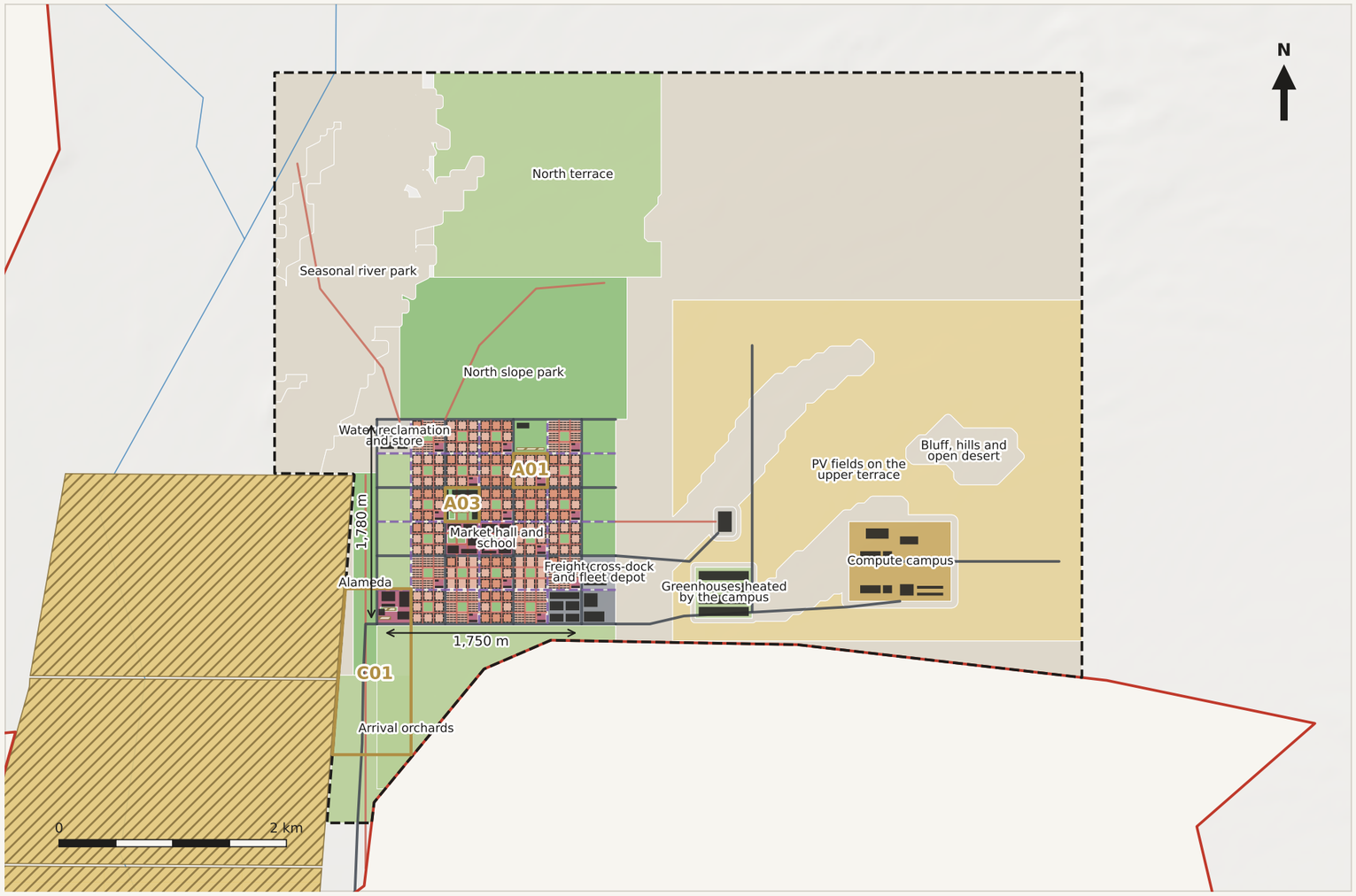
Terrace Oasis: a city that grows its own shade beside the first fields

The same node IDs are used on the plans, images and technical evidence

Whole-site location of the master design area



Master-area enlargement: selected nodes and movement networks



Network colours: people / salmon; machines / charcoal; shared / purple. Follow the submitted networks; no extra direct links are invented.

C01 | Entrance and arrival sequence

south-west of the master-area centre, 3.80 km; node area 92.65 ha. See the shared locator for access and connections.

Role and connections

Everyone arrives the same way: 3.5 km beside the agrivoltaic fields the city grew out of, then under one canopy where the coach stops and the city begins on foot. If arrival is a car park, the city has lied about what it is.

A01 | A future housing prototype

south-west of the master-area centre, 1.59 km; node area 9.00 ha. See the shared locator for access and connections.

Role and connections

Housing is where heat, earthquakes, water and robots meet every day. If a family can live here in January and in a Zonda week without a car or an air-conditioner running all day, the rest of the city follows.

A03 | Public space in heat and Zonda wind

south-west of the master-area centre, 2.26 km; node area 9.00 ha. See the shared locator for access and connections.

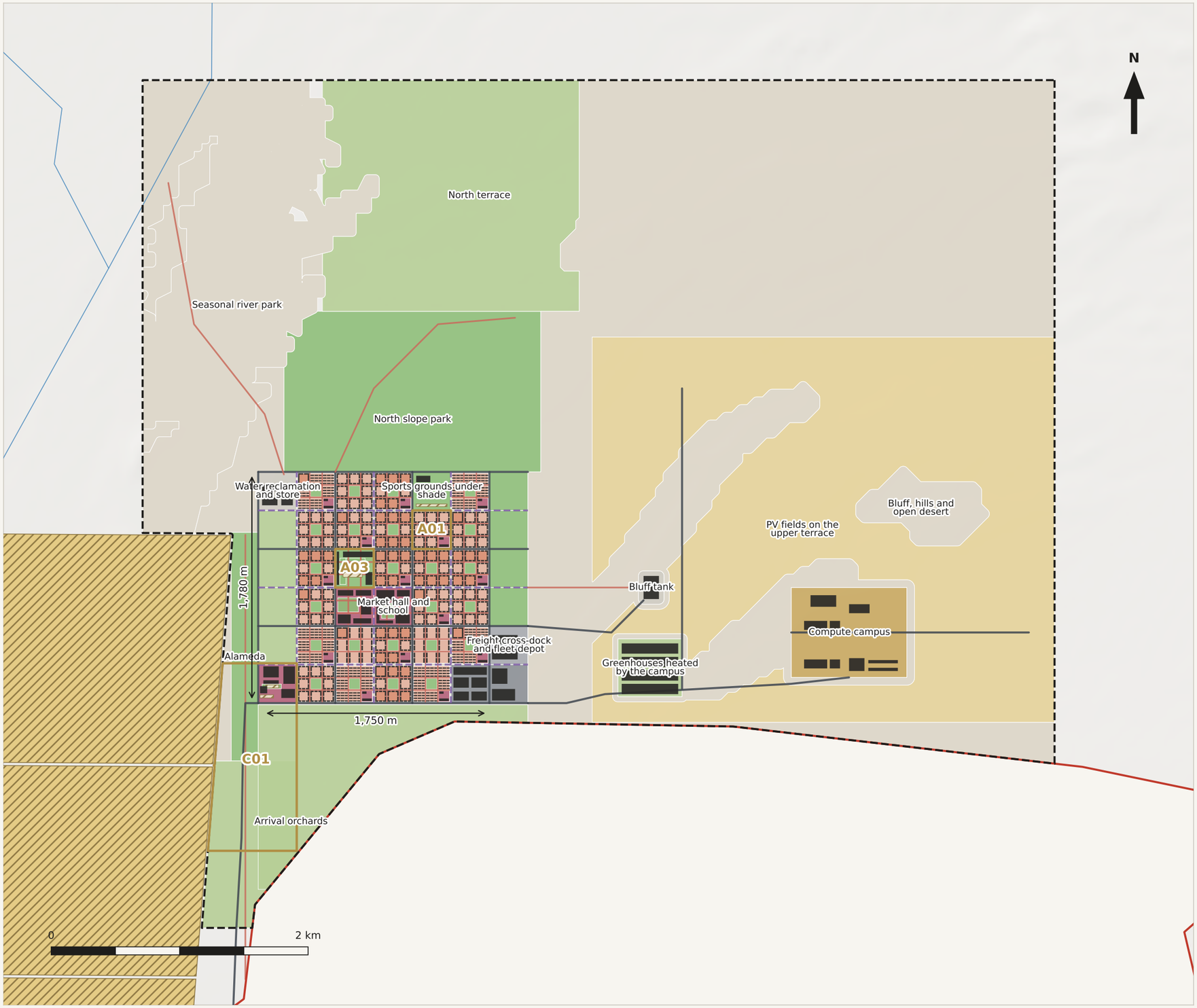
Role and connections

A desert city needs one room where everyone can be outside at once. Here it has to work at 45 C and in a hot north-west wind, with water that is never wasted.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Scale 2 · figure-ground and land use



Legend

- Agriculture
- Civic
- Compute
- Energy
- Industry
- Infrastructure
- Logistics
- Mixed use
- Open space
- Reserve
- Residential
- Buildings (figure-ground)
- Master design area
- Key node
- Seasonal channel

Master design area

36 km² · drawn by the entrant

Recomputed

Site coverage 2.90% · FAR 0.06 · GFA 2,309,984 m²

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Floors from buildings.geojson; metrics recomputed



Legend

- 1 floor
- 2-3 floors
- 4-6 floors
- 7+ floors
- Master design area

Land use in the master area (ha)

reserve 1,666
energy 782
agriculture 586
open space 361
residential 102
compute 63
mixed use 38
civic 38
industry 18
infrastructure 14
logistics 9.01

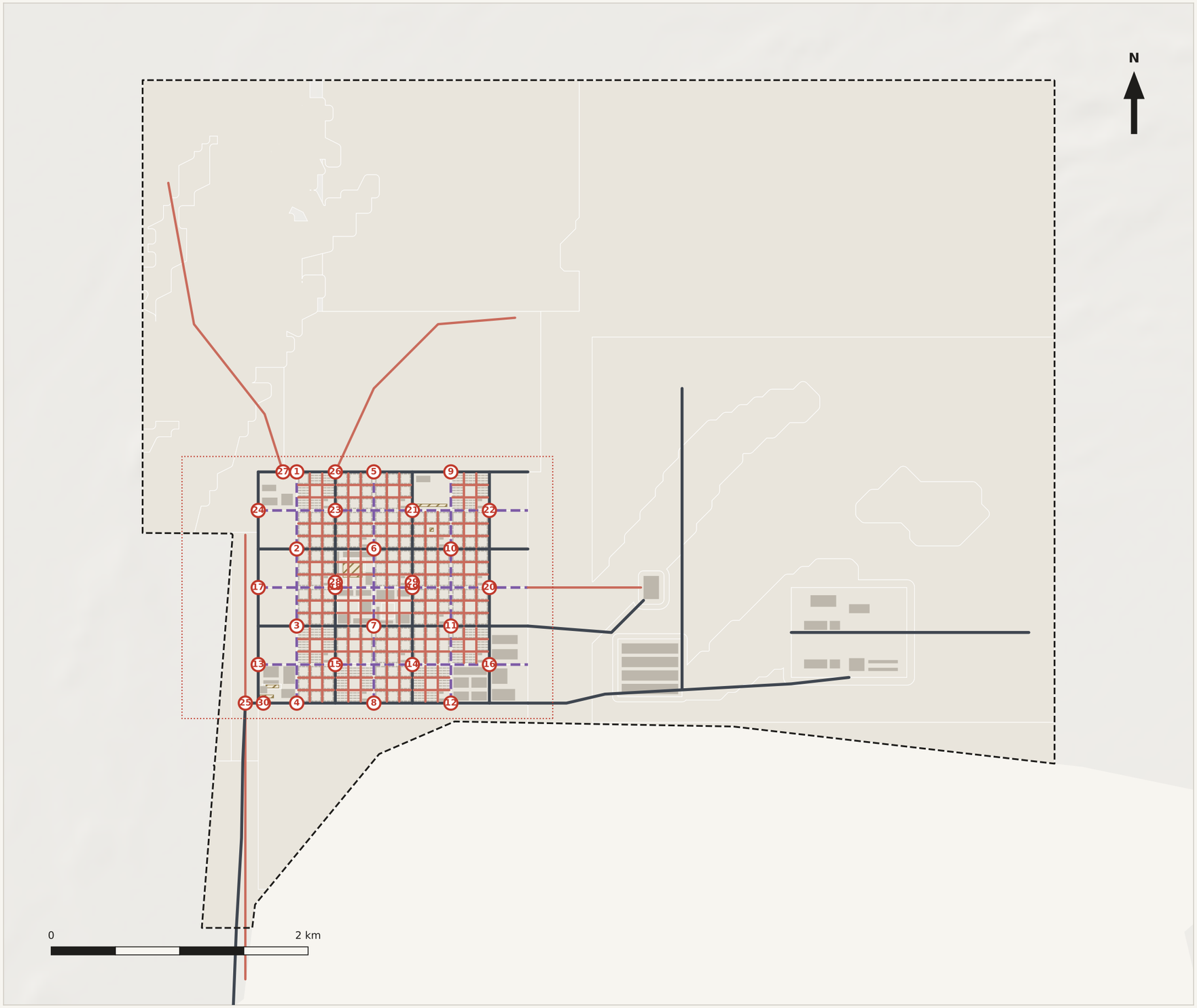
Recomputed from geometry

Coverage 2.90% · FAR 0.06 · open space 9.90%

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Every human-machine interface numbered



Legend

- People 40 km
- Machines + goods 30 km
- Shared 12 km
- Human-machine interface (30)

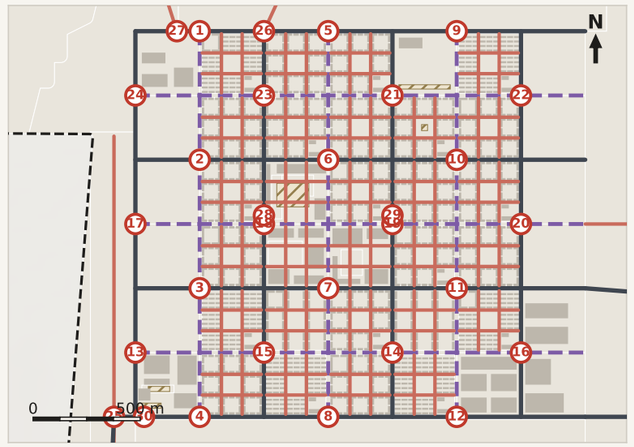
Machine traffic rule

combined

Interfaces

30 at-grade points where people and machines meet

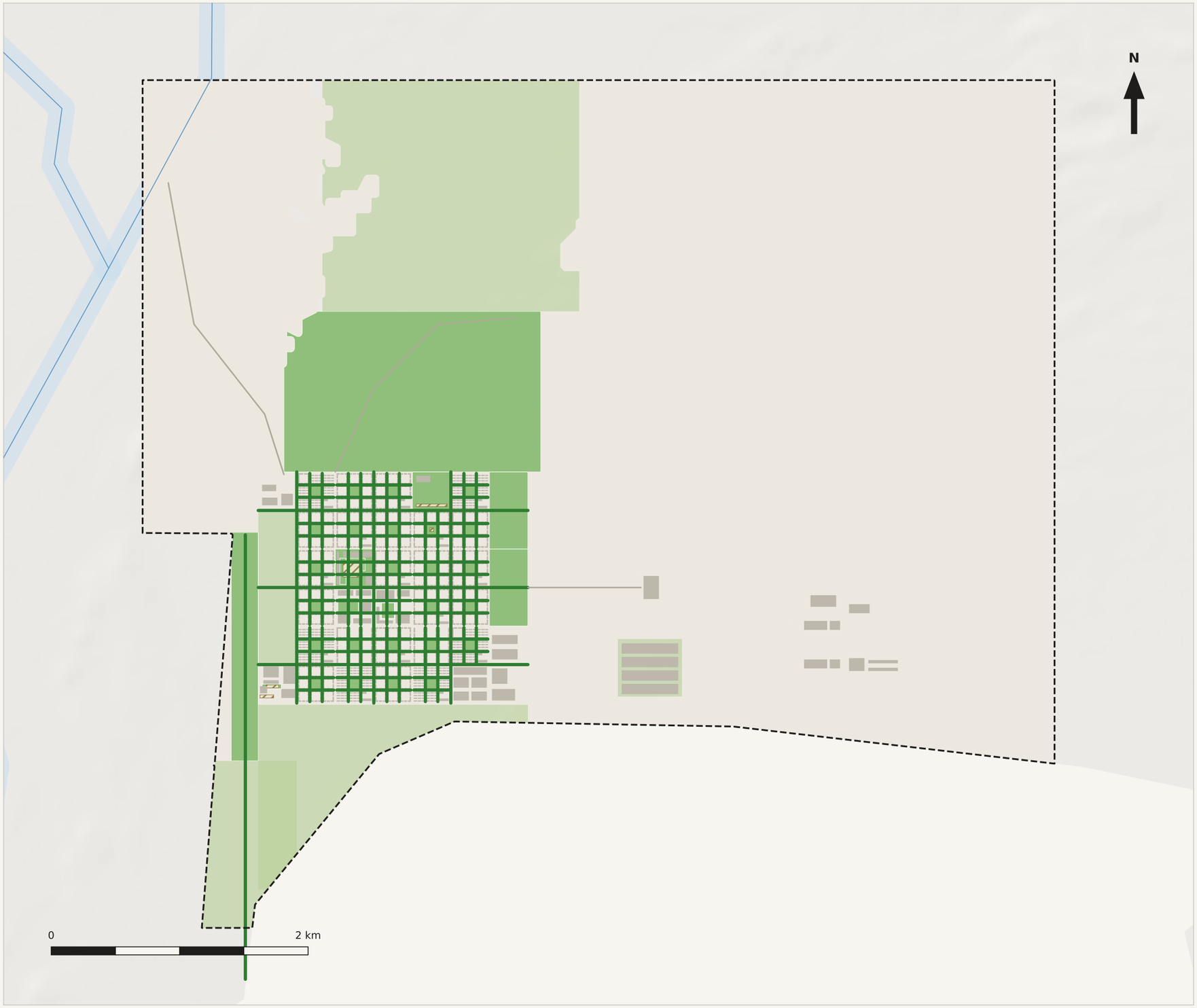
Interfaces, enlarged



Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Channels are DEM-derived flow lines



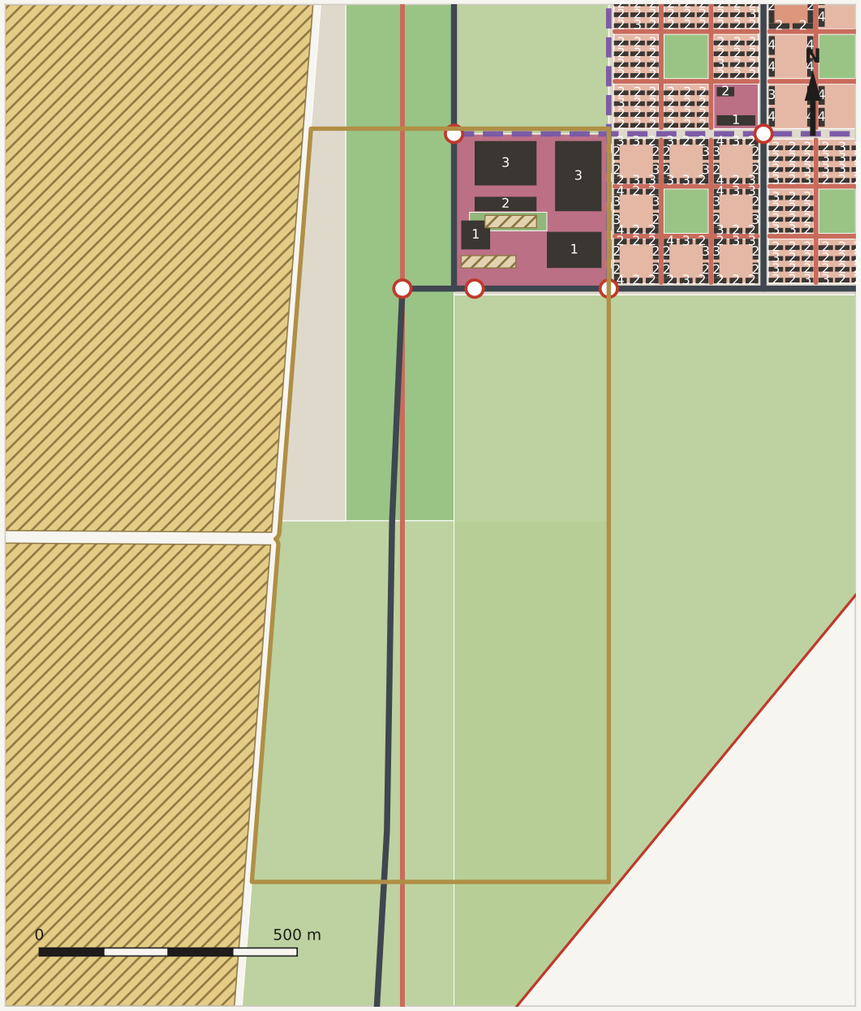
Legend
- Open space 9.90%
- Agriculture
- Shaded walking 90%
- Unshaded walking
- Seasonal channel
- Flood corridor shown at 250 m / 100 m

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Location: south-west of the master-area centre, 3.80 km; node area 92.65 ha. See the shared locator for access and connections.

Plan · floors marked on each building · interfaces circled



Arriving along the first fields

Everyone arrives the same way: 3.5 km beside the agrivoltaic fields the city grew out of, then under one canopy where the coach stops and the city begins on foot. If arrival is a car park, the city has lied about what it is.

Arriving along the first fields



C01-aerial · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view C01-aerial · illustrates the geometry, not evidence

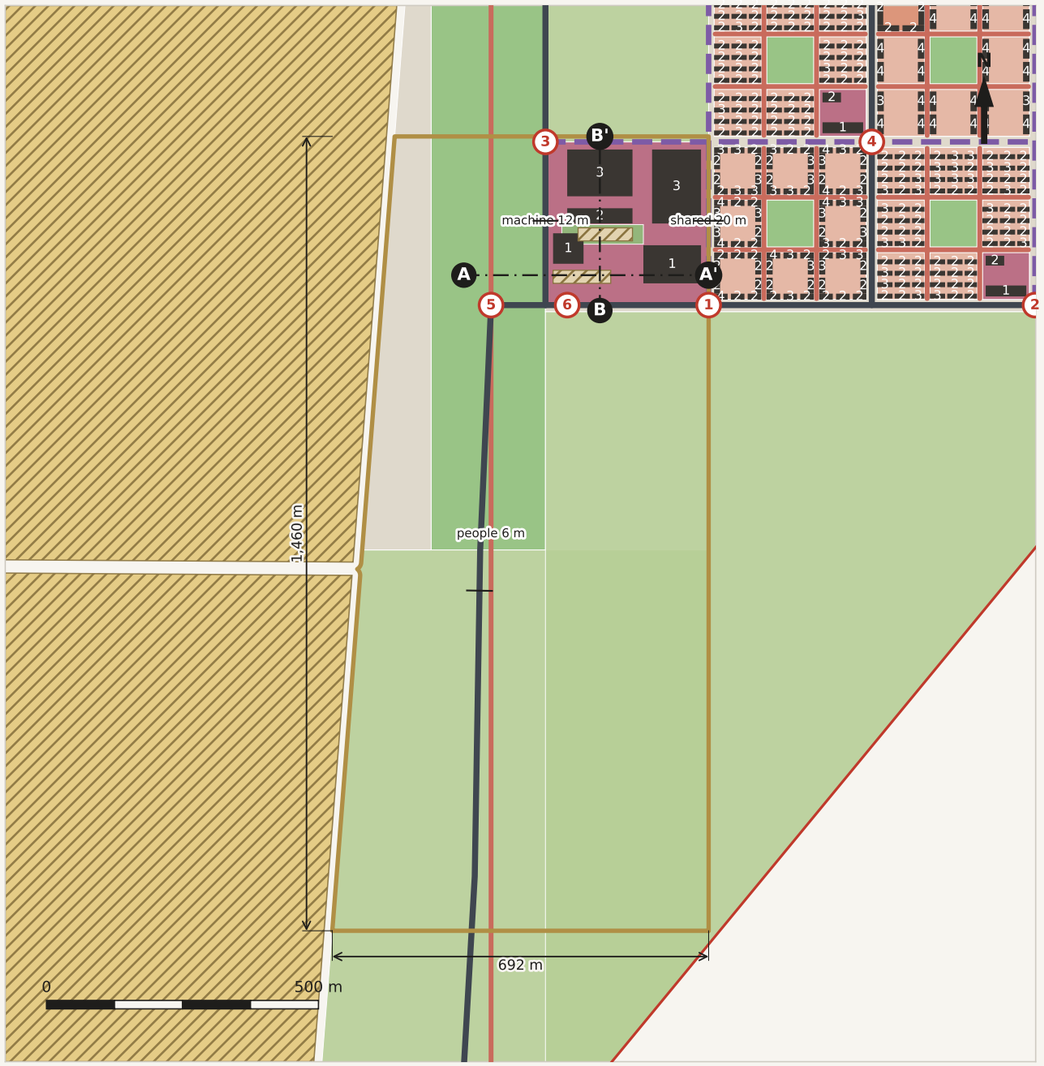


C01-eye-day · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view C01-eye-day · illustrates the geometry, not evidence

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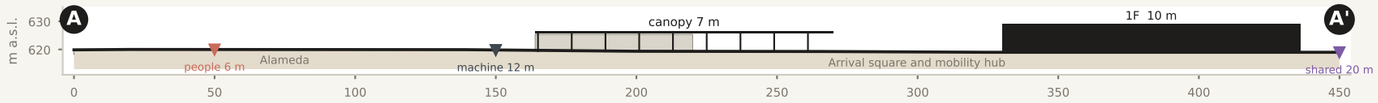
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Plan · overall dimensions, street widths, interfaces numbered, section lines

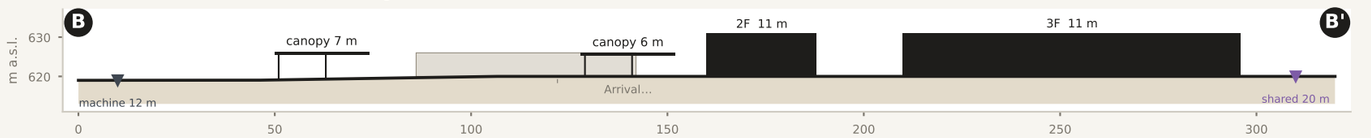


Scale 3 · dimensions and sections from the geometry

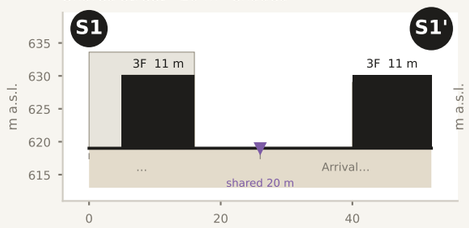
Section A-A' · 450 m · true scale, looking left of the cut



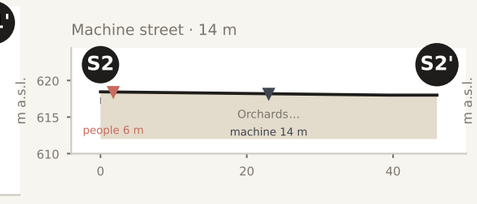
Section B-B' · 320 m · true scale, looking left of the cut



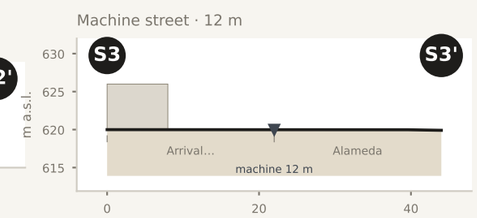
Shared street · 20 m · shaded



Machine street · 14 m



Machine street · 12 m



Numbers in the node · recomputed from the geometry

Node area 926,500 m² (93 ha)
Buildings 5 · footprint 36,480 m² · floor area 84,960 m²
Coverage 3.90% · FAR 0.09 · open space 18%
Floors: 2 x 1F, 1 x 2F, 2 x 3F
Floor area by use: mixed use 67,700 m², civic 14,100 m², infrastructure 3,100 m²
Canopies (roof on posts, no floor area) 4,940 m²
Networks in the node: machine 1,861 m, people 1,460 m, shared 610 m
Shaded share of the walking network 100%
Human-machine interfaces 4

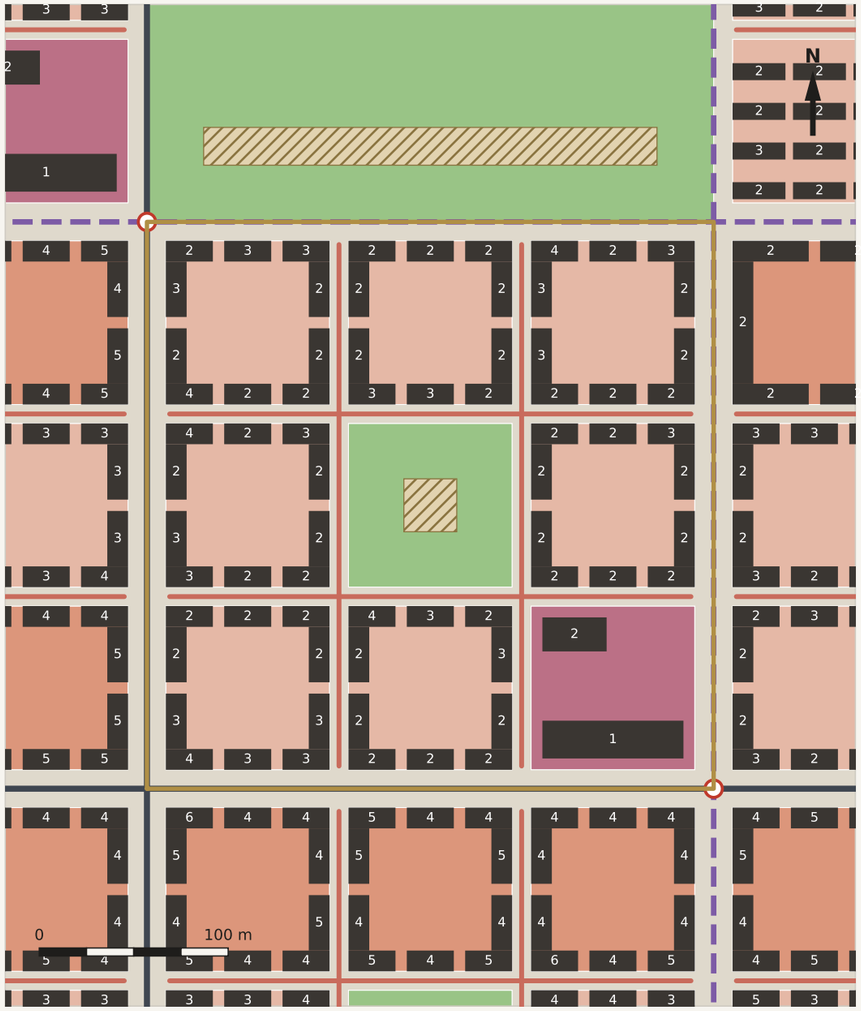
Cut buildings in black (floors x 3.5 m, or height_m); beyond the cut, buildings and trees within 45 m in grey; trees and people as placed in the 3D scene; terrain from the DEM.

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Terrace Oasis: a city that grows its own shade beside the first fields

Location: south-west of the master-area centre, 1.59 km; node area 9.00 ha. See the shared locator for access and connections.

Plan · floors marked on each building · interfaces circled



A superblock of courtyard houses

Housing is where heat, earthquakes, water and robots meet every day. If a family can live here in January and in a Zonda week without a car or an air-conditioner running all day, the rest of the city follows.

A superblock of courtyard houses



A01-aerial · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view A01-aerial · illustrates the geometry, not evidence

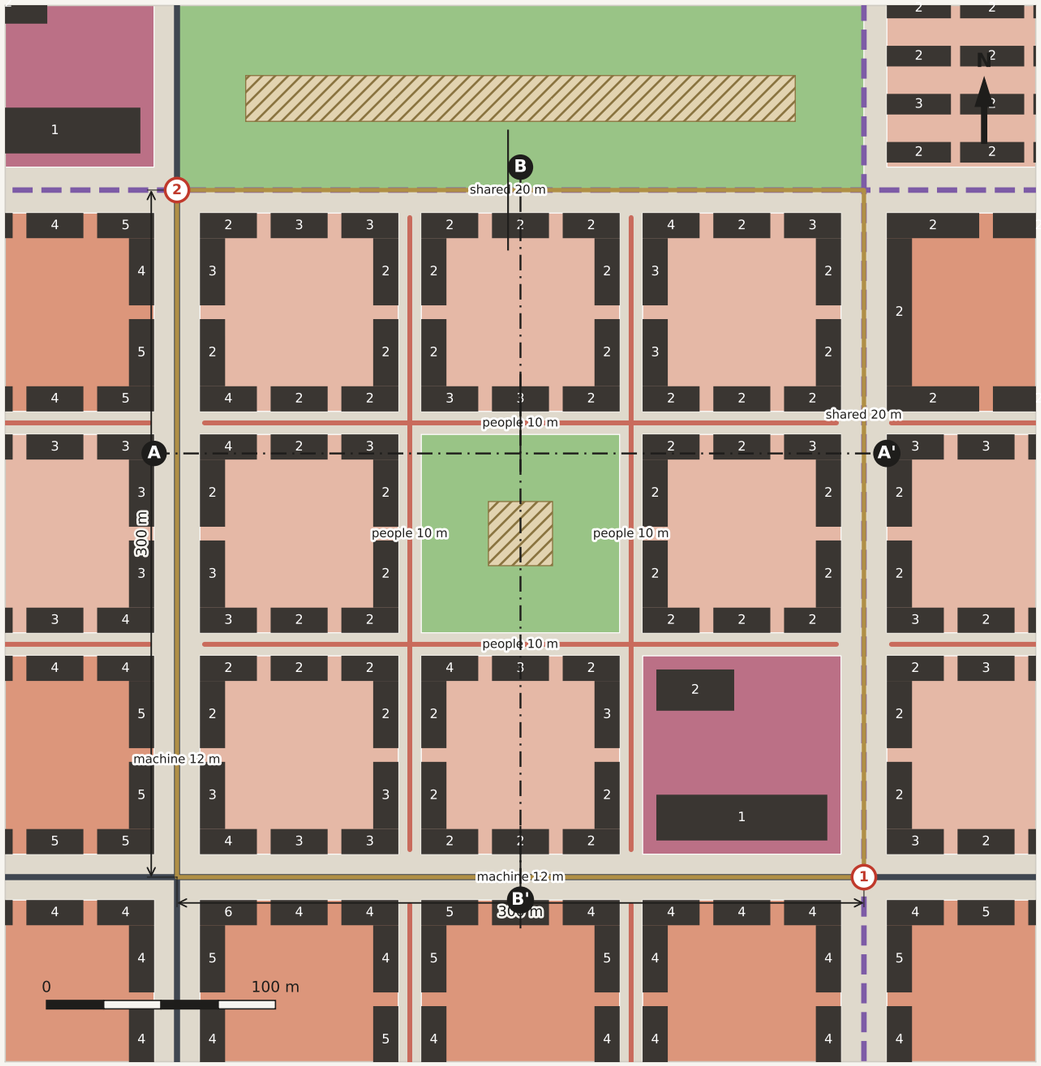


A01-eye-day · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view A01-eye-day · illustrates the geometry, not evidence

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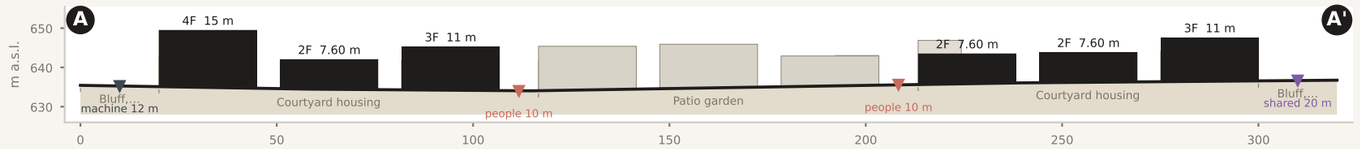
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Plan · overall dimensions, street widths, interfaces numbered, section lines

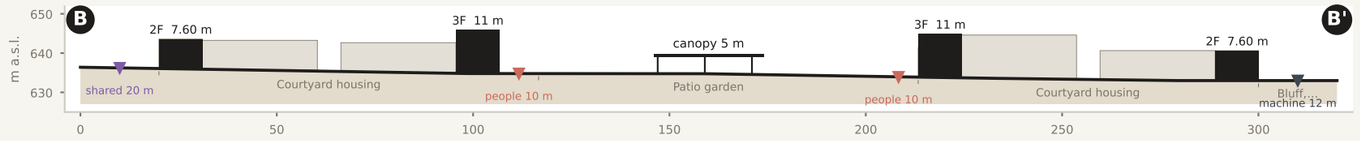


Scale 3 · dimensions and sections from the geometry

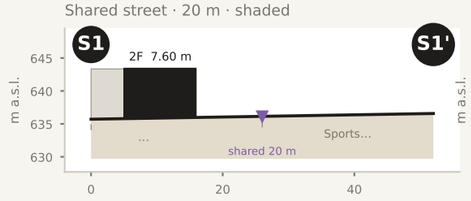
Section A-A' · 320 m · true scale, looking left of the cut



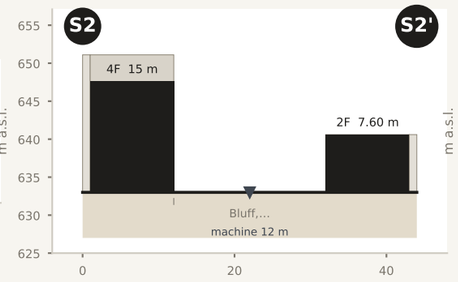
Section B-B' · 320 m · true scale, looking left of the cut



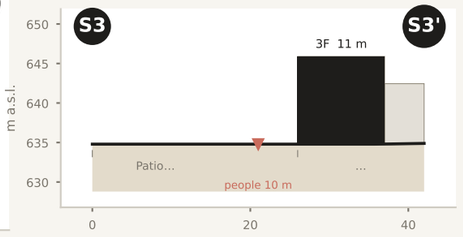
Shared street · 20 m · shaded



Machine street · 12 m



People street · 10 m · shaded



Numbers in the node · recomputed from the geometry

Node area 90,000 m² (9 ha)
Buildings 72 · footprint 22,640 m² · floor area 51,790 m²
Coverage 25% · FAR 0.58 · open space 8.30%
Floors: 1 x 1F, 48 x 2F, 18 x 3F, 5 x 4F
Floor area by use: residential 49,100 m², civic 2,700 m²
Canopies (roof on posts, no floor area) 780 m²
Networks in the node: machine 403 m, people 1,104 m, shared 486 m
Shaded share of the walking network 100%
Human-machine interfaces 2

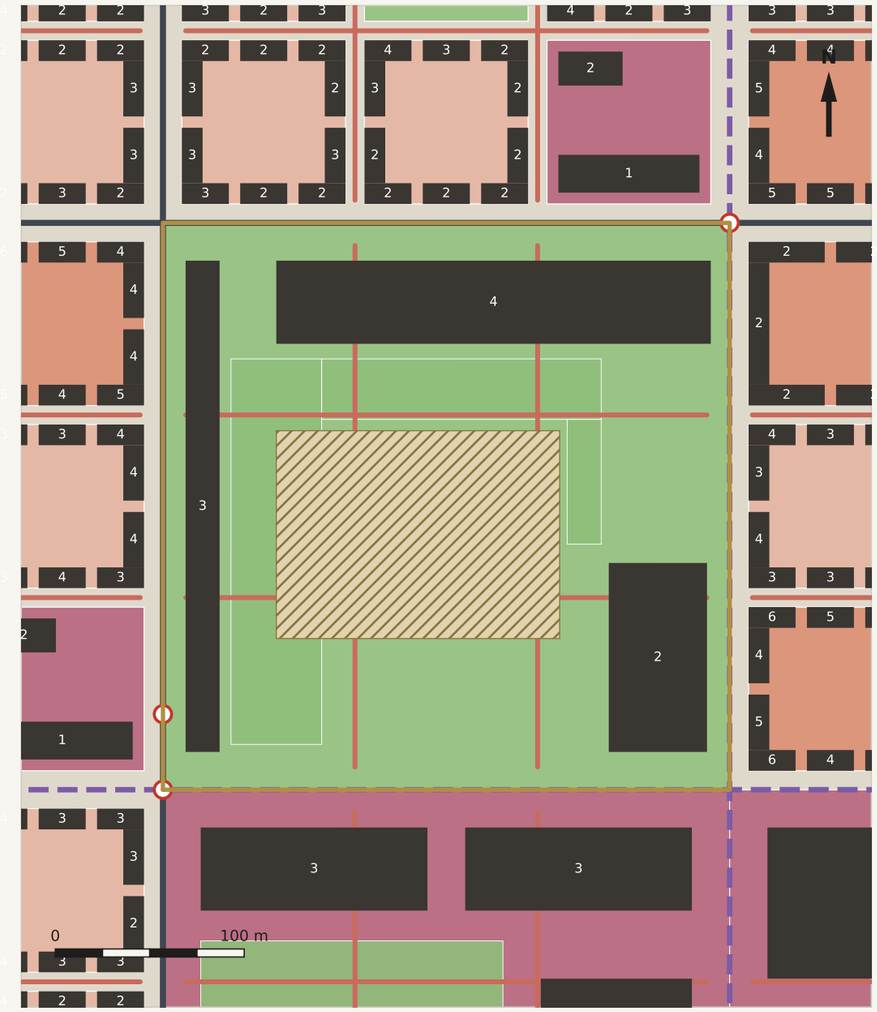
Cut buildings in black (floors x 3.5 m, or height_m); beyond the cut, buildings and trees within 45 m in grey; trees and people as placed in the 3D scene; terrain from the DEM.

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Plan · floors marked on each building · interfaces circled



Plaza del Agua: a square for 45 degrees and the Zonda

A desert city needs one room where everyone can be outside at once. Here it has to work at 45 C and in a hot north-west wind, with water that is never wasted.

Plaza del Agua: a square for 45 degrees and the Zonda



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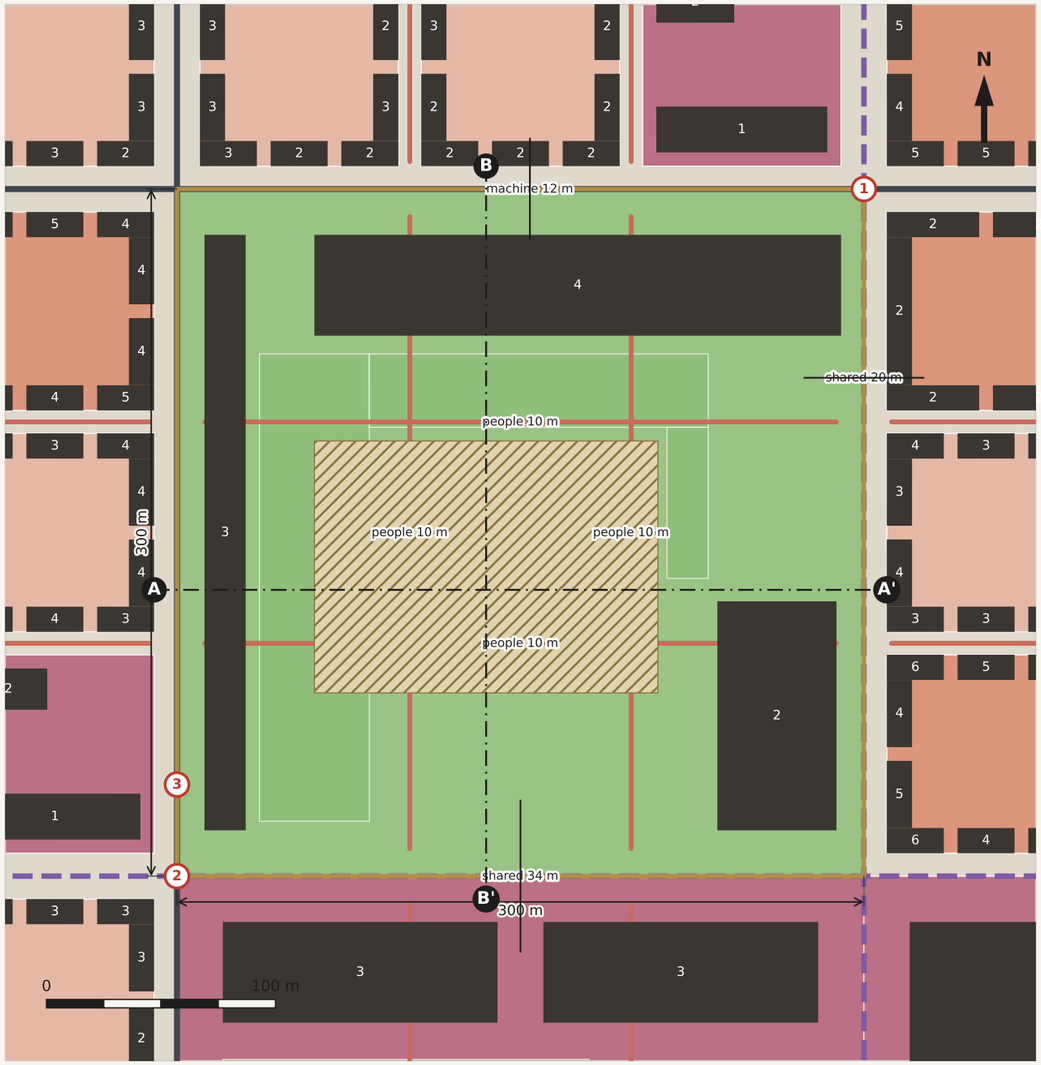


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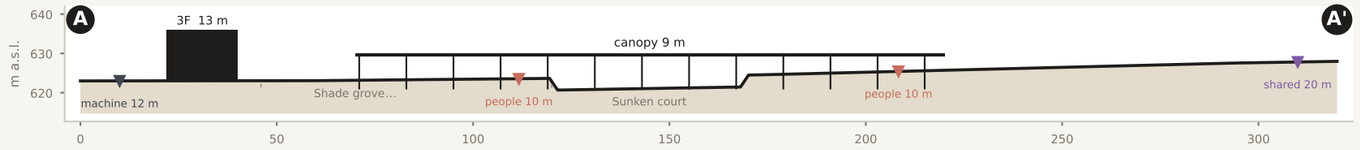
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Plan · overall dimensions, street widths, interfaces numbered, section lines

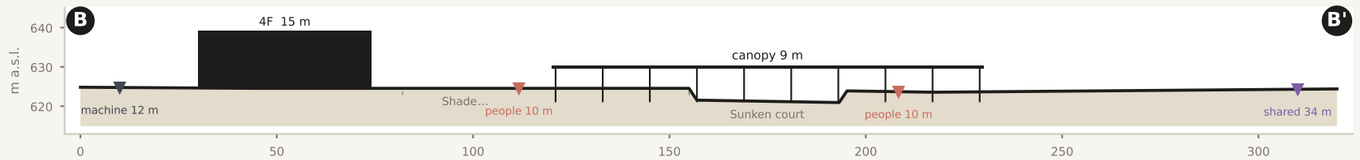


Scale 3 · dimensions and sections from the geometry

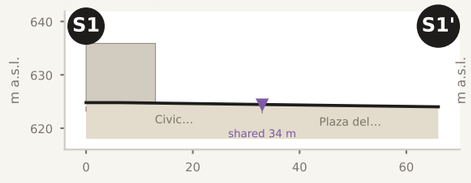
Section A-A' · 320 m · true scale, looking left of the cut



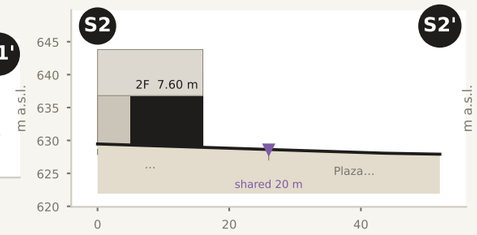
Section B-B' · 320 m · true scale, looking left of the cut



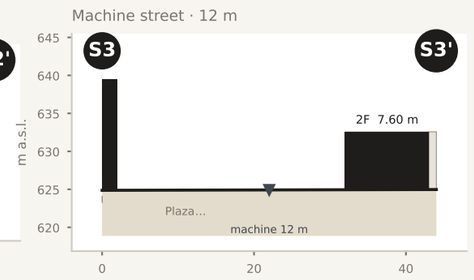
Shared street · 34 m · shaded



Shared street · 20 m · shaded



Machine street · 12 m



Numbers in the node · recomputed from the geometry

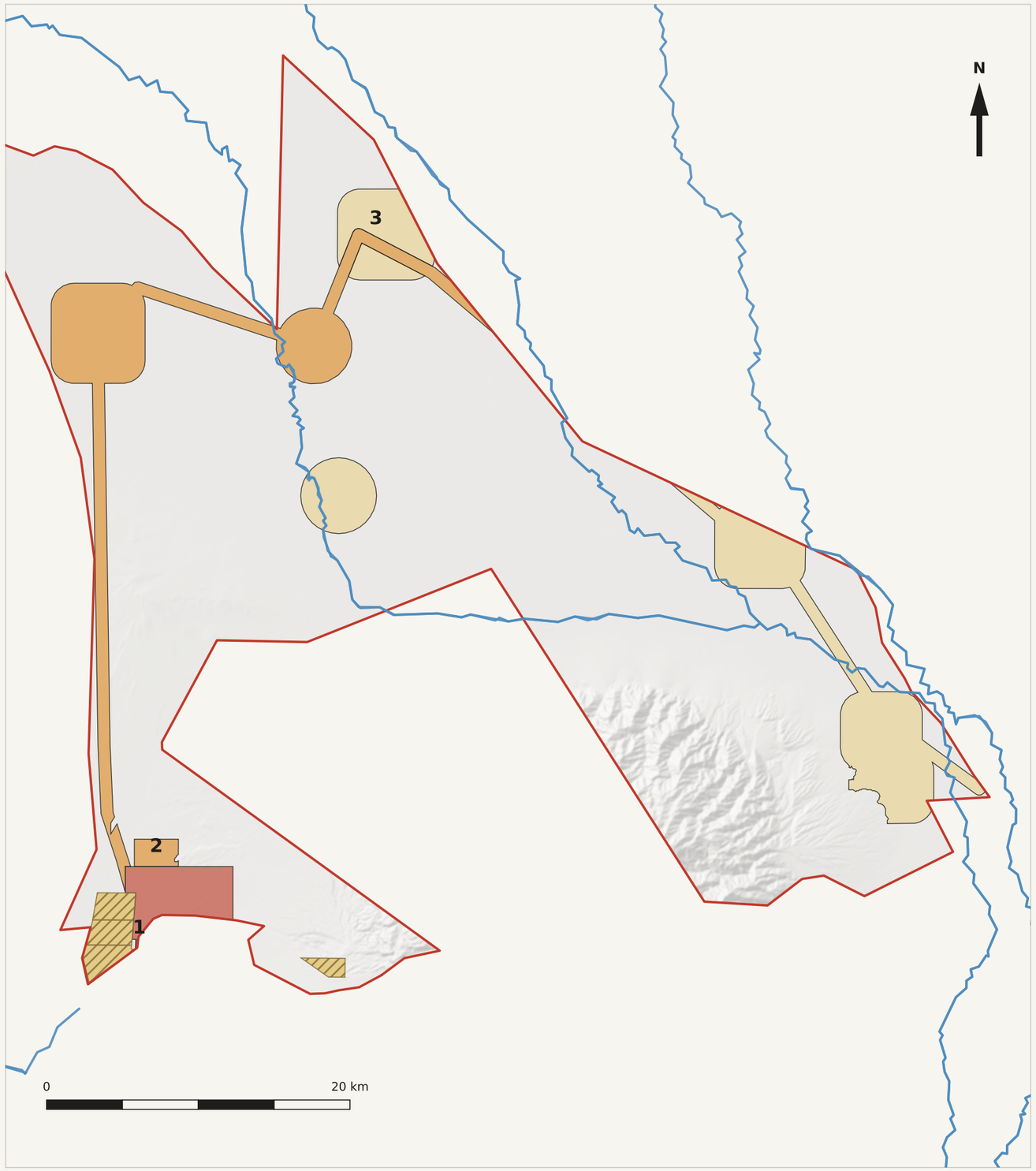
Node area 90,000 m² (9 ha)
Buildings 3 · footprint 20,000 m² · floor area 64,930 m²
Coverage 22% · FAR 0.72 · open space 100%
Floors: 1 x 2F, 1 x 3F, 1 x 4F
Floor area by use: mixed use 40,500 m², civic 24,400 m²
Canopies (roof on posts, no floor area) 16,500 m²
Networks in the node: machine 292 m, people 1,104 m, shared 465 m
Shaded share of the walking network 100%
Human-machine interfaces 3

Cut buildings in black (floors x 3.5 m, or height_m); beyond the cut, buildings and trees within 45 m in grey; trees and people as placed in the 3D scene; terrain from the DEM.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Observable triggers, not dates



Legend

- Phase 1 24 km²
- Phase 2 109 km²
- Phase 3 122 km²

Phase 1 - Terrace Oasis: the first city, its fields, PV and campus, and the arrival road

Phase one: no trigger.

Phase 2 - Upper town on the north terrace; the Line north to Oasis 2; second campus

Lower terrace above 85% occupied for 12 consecutive months AND wellfield drawdown below 0.5 m a year for 24 months AND a second abstraction licence granted AND the open-data office has published 24 monthly water and power accounts

Phase 3 - East: Oasis 3 and 4, port of entry, gateway industry, the Line to RP 510

Access corridor to RP 510 financed AND committed export throughput above 40,000 t a year AND phase-two population above 40,000

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Full accounts and checks are in the separate technical appendix

Indicator	Value	Unit	Scope / phase	Evidence
Master design area	36	km2	Master area	computed
Residents	41,340	persons	Phase 1	declared
Jobs	7,891	jobs	Phase 1	declared
Homes	17,224	units	Phase 1	declared
Floor area ratio	0.06	ratio	Master area	computed
Open space	9.91	%	Master area	computed
Annual water demand	3,599,292	m3/yr	Phase 1	assumed
Compute load	40	MW	Phase 1	declared
Hourly clean energy match	100	%	Phase 1	assumed
Water reuse	90	%	Phase 1	assumed
Waste heat reuse	9	%	Phase 1	assumed

Reading these figures

The whole site is 1,656 km²; master-area figures are not whole-site totals. Phase 1 means full build-out and occupancy unless stated otherwise. Evidence tags describe the basis, not approval. Full system land accounts, assumptions and calculations remain in the appendix and metrics.json.

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